



EMPLOYEE PERFORMANCE & PAYROLL MANAGEMENT SYSTEM

Guided by **SAYAN SAU SIR(COE)** and done by MOUSOM KOLEY & ARNIK JANA (student of COE) a comprehensive Excel Dashboard for HR Analytics, Performance Tracking and Payroll Intelligence across 6 Departments & 350 Employees

350

Employees

225

Active

6

Departments

10+

KPIs Tracked

THE PROBLEM

Why This System Was Needed

✗ Messy, Inconsistent Raw Data

Names in ALL CAPS or lowercase, gender as M/F shortcuts, mismatched date formats (text vs serial), negative attendance values, blank salary cells — 350 records filled with entry errors.

✗ No Unified Performance View

Performance scores, ratings, and attendance lived in separate rows with no summary. Management had no quick way to compare departments or flag low performers.

✗ Payroll Opacity

Basic Salary, HRA, Allowances, Overtime and Deductions were unvalidated. Net salary totals per department were unavailable without manual effort.

✗ Zero Attrition Visibility

With Active, Resigned and On Leave statuses mixed together, tracking headcount health and calculating attrition rates required tedious manual counting.



IMPACT WITHOUT THE SYSTEM



Hours wasted

per payroll cycle cleaning data manually



Risk of errors

in salary calculations due to dirty inputs



No attrition alerts

management reacted too late to resignations



Dept comparisons

impossible without summarized data



Audit failures

inconsistent formats fail data validation

OUR SOLUTION

How This Excel System Solves It

01 Data Cleaning Layer

- ✓ PROPER() / UPPER() / TRIM() to fix name & dept casing
- ✓ IF(OR()) to expand M/F → Male/Female
- ✓ ABS() to fix negative attendance values
- ✓ TEXT() + DATEVALUE() to unify date formats
- ✓ IFERROR(VALUE,0) to handle blank salary cells

02 KPI Dashboard

- ✓ COUNTIFS() for Active/Resigned/On Leave headcount
- ✓ SUMIFS() for department-level payroll totals
- ✓ AVERAGEIFS() for performance score by dept
- ✓ Attrition Rate = $\text{Resigned} \div \text{Total} \times 100$
- ✓ Visual KPI cards with department breakdowns

03 Interactivity & Slicers

- ✓ Slicers on Department, Status, Gender & Rating
- ✓ One-click filtering of all charts simultaneously
- ✓ Dept Summary sheet with bar charts per dept
- ✓ Pivot tables behind every chart for live updates
- ✓ Clean navigation via Cover sheet with hyperlinks

Data Issue	Raw Data Example	Formula Applied	Clean Result
Name Casing	KAVITA, arun Shah	<code>PROPER(TRIM(B2))</code>	Kavita, Arun Shah
Dept Casing	finance, FINANCE	<code>UPPER(TRIM(E2))</code>	FINANCE
Gender Abbrev.	M, F, Female	<code>IF(OR(D2="M", "m"), "Male", ...)</code>	Male / Female
Neg. Attendance	-24, -15, -22	<code>ABS(O2)</code>	24, 15, 22
Date Formats	01/5/2019, 43045	<code>TEXT(DATEVALUE(G2), "DD-MMM-YYYY")</code>	05-Jan-2019
Missing Salary	(blank)	<code>IFERROR(VALUE(I2), 0)</code>	0

DATA QUALITY
RESULT

350

Records
Cleaned

6

Issues
Fixed

100%

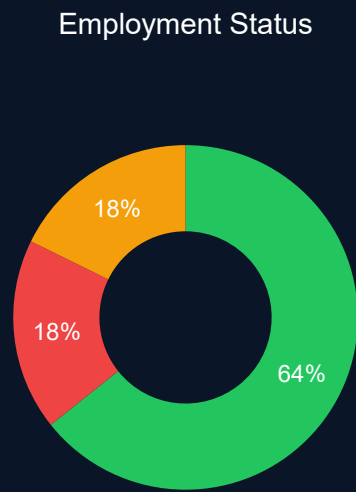
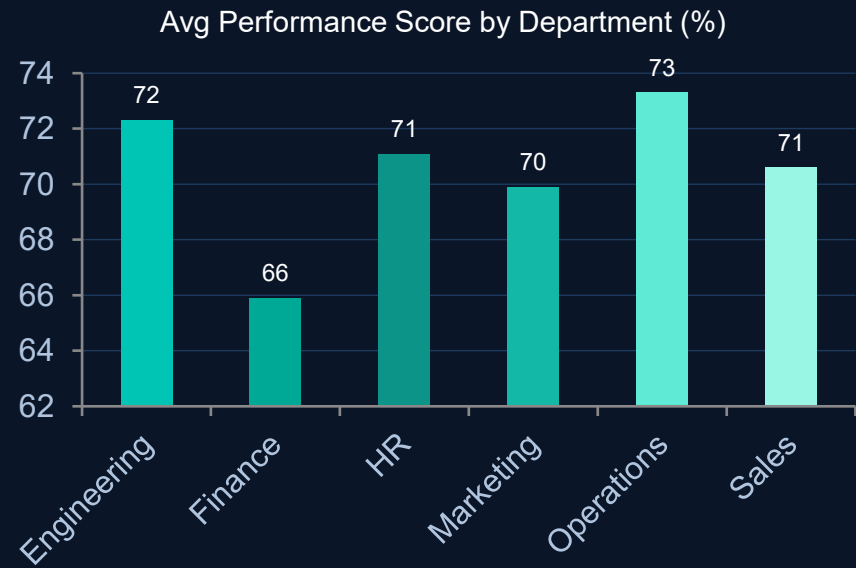
Formula
Driven

0

Manual
Edits

KPI DASHBOARD

15+ Metrics at a Glance



CHARTS & SLICERS IN THIS PROJECT

Interactive Visual Intelligence



CHARTS USED

Clustered Column Chart

Department-wise Avg Performance Score comparison. X-axis = Department, Y-axis = Score %

Doughnut / Pie Chart

Employment Status split (Active vs Resigned vs On Leave) shown as % share of total headcount.

Horizontal Bar Chart

Total Payroll (₹) per department — easy left-to-right reading for salary volume comparison.

Scatter / Data Points

Performance score vs Attendance Days to identify high-performers with poor attendance.



WHY THIS SLICER?

A Slicer is a visual Excel filter button panel linked to Pivot Tables. Clicking a slicer button instantly filters all connected charts and tables — no formulas needed.

SLICERS USED IN THIS PROJECT:

Department

Filter by Engineering, Finance, HR, Marketing, Operations, Sales

Employment Status

Toggle between Active / Resigned / On Leave views

Gender

Compare Male vs Female performance and pay

Performance Rating

Drill into Poor, Average, Good, Excellent employees

PROJECT SUMMARY & IMPACT

What This System Delivers

✓ WHAT WAS BUILT

- ▶ 5-sheet Excel workbook with full navigation
- ▶ Raw Data → Clean Data pipeline using 10 Excel functions
- ▶ KPI Dashboard: 7 headline metrics + department breakdowns
- ▶ 4 interactive charts (Column, Doughnut, Bar, Scatter)
- ▶ 4 slicers: Department, Status, Gender, Performance Rating
- ▶ Dept Summary sheet with per-department bar charts
- ▶ Instructions sheet with formula guide & KPI definitions

Built entirely in Microsoft Excel · No external tools required

📊 BUSINESS IMPACT

100%

Data Accuracy
after cleaning

18%

Attrition Rate
now visible instantly

4.11Cr

Total Payroll
now tracked

70.4%

Avg Performance
benchmark established